Joseph Wardle

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Work Experience

Pipeline TD - 'Sandwich Kwon Do' and 'Honey Business' October 2025 — Present
Brigham Young University — Animation Department Capstone Films *Provo, UT*

- **Maintain, debug, and extend** a film-scale, OS-agnostic **USD** pipeline used by **50+ artists** across **Linux** and Windows.
- Craft **Qt** (PySide2) artist tools across **Maya**, **Houdini**, **Nuke**, and **Substance Painter**; integrated with **ShotGrid** for asset registration, version tracking, and automatic playblasts for dailies.
- Provide front-line support to production artists by triaging tool bugs, fixing broken publishes, and **wrangling shot and asset data**, including introducing **version control**.
- Built a telemetry system harvesting per-frame render times and memory from **Husk** and **RenderMan** via **Pixar Tractor** across a **107-node** farm.
- Engineered a unified publish where **Maya** and **Substance Painter** invoke a **Houdini** subprocess to assemble a **USD** asset.
- Built proxy-generation tools for point-instanced **USD** assets (foliage), **improving viewport performance 70x**.

Web Developer March 2025 — November 2025
Brigham Young University — Computer Science Department *Provo, UT*

- Engineer and maintain department web services as **Django** applications using SWIFT practices.
- Design REST endpoints and internal admin UIs for non-technical users.
- Build **GitHub CI/CD pipelines** for test, image build, and deploy; local dev with **Docker Compose**;
- Project tracking via **Jira**

Audio Engineer September 2023 — November 2025
Brigham Young University *Provo, UT*

- Sound-system design, streaming, live audio mixing for stadium sports, large-scale devotionals, and performances.

Projects

Javelin - Real-Time Rigid Body Physics Engine (github.com/joseph-wardle/javelin) 2023 — Present

- Build a rigid body simulator from scratch in **C++23**; handles **5,000+ concurrent dynamic bodies** at 60 Hz.
- Multithreaded broad phase (dynamic/static BVH), SAT + GJK narrow phase with contact manifolds, warm-started projected Gauss-Seidel solver; Tracy integration for per-tick diagnostics, profiling, and telemetry.
- Implemented in **C++23 named modules** throughout; custom math library (vec3, mat3, mat4, quat); lock-free atomics for manual step control from the render thread. **ImGui** used for runtime developer UI.

Real-Time Ray Tracer (github.com/DallinClark/RealTimeRaytracer) March 2025 — August 2025

- Built a real-time ray tracer using **Vulkan 1.3** hardware-accelerated ray tracing; authored pipeline/bootstrap code.
- Responsible for pipeline setup, descriptor management, development tooling, and custom memory allocators.

Film Grain Synthesis (github.com/joseph-wardle/film_grain) August 2025

- Implemented physically motivated film grain algorithms (Newson et al., 2017) in **Rust** with **WGPU** compute shaders;
- Up to **160X** faster than the C reference on CPU; **68x** faster on GPU for large pixel-wise workloads.

Education

Brigham Young University Provo, UT
Bachelor of Science, Computer Science (Animation & Games Emphasis) *Aug 2023 — Expected May 2027*

- Relevant Coursework: Data Structures, Multithreading, Linear Algebra, Modeling, Rigging, Shading, FX (3.95 GPA)

Skills

- **Languages and Systems:** Python, MEL, C++23, Rust, TypeScript, Bash, Java, C#, HTML/CSS
- **Systems:** Linux, Windows, systemd/Quadlet, Kubernetes, Podman/Docker, NGINX/Caddy
- **CI/CD & Deploy:** GitHub Actions, Jenkins, TeamCity, Git, Perforce
- **Render & Pipeline:** USD, Pixar Tractor, Houdini, Maya, Nuke, ShotGrid, MoonRay
- **Other Software:** Unreal Engine, Unity, Adobe, Jira, Confluence